FAKULTA INFORMATIKY A INFORMAČNÝCH TECHNOLÓGIÍ SLOVENSKÁ TECHNICKÁ UNIVERZITA

Ilkovičova 2, 842 16 Bratislava 4

PIS - Coffee shop

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Subject: Principles of information systems

Exercise: Wednesday 14:00

Lecturer: Ing. Fedor Lehocki, PhD.

Created by: Erik Mokrán, Vadym Tilihuzov, Sira Dariia, Nedbal Filip

Description of the process

Processes

1 Before the coffee shop opens

Manager, Waiter, and Barista arrives at work

Manager

- The manager checks the condition of the equipment.
- If the equipment is functioning properly, he prepares the cash register. If not, he either calls the repair service or schedules such a call for a specific date.
- If additional products need to be ordered, the manager places an order for a specific date.
- (Optional, can be done in 2 processes (when the coffee shop opens)) If an order (products) has arrived, the manager signs the documents confirming the receipt of the order.

Barista

- The barista prepares their workstation.
- The barista checks the coffee equipment and cleans it.
- They also check the inventory of ingredients. If they see that it needs to be reordered soon, they proceed to the next step. If not, they notify the manager of the need to place an order.

Waiter

- The waiter checks the condition of the dining area and clears tables if necessary.
- They also check for clean dishes and utensils for service.

- (Optional, can be done in 2 processes (when the coffee shop opens)) Helps unload the delivery truck. Afterward, either they display it prominently for customers or bring it to the barista.

- Once all preparatory work is completed, the coffee shop is ready to open and serve customers.

2 The coffee shop is open

- A customer enters the coffee shop.
- The customer approaches the cashier.
- The manager/waiter greets the customer. The customer may either go straight to the counter to order something or be greeted by the waiter/manager and offered a table.
    - Option a - The customer chooses a table, then they are escorted to the table. The customer can place an order immediately, and then the waiter/manager takes the order. This, in turn, prompts them to go to the cashier and leave the order with the barista. The barista starts preparing drinks according to the order. After completion, if required, the drinks are brought to the customer. If the order requires the barista to prepare something, it will only be brought when the order is ready. This also includes the order taker checking if all items from the menu are available or not. If some items are not available, an alternative is suggested. At the end, when the customer wants to leave, they call the waiter/manager again, pay, and then leave. The manager/waiter processes the payment and issues a receipt to the customer.
    - Option b - The customer may decline a table and go to the barista (order counter), queue if necessary, where the order discussion will start. Here, it will also be checked if all products are available. An important difference is that the customer pays immediately for the order, unlike in option a, where the customer pays when they are ready to leave.

Common: Examples of what the barista may prepare:

- Brews tea, prepares hot chocolate.
- Prepares coffee drinks (espresso, cappuccino, latte):
    - Grinds coffee beans.
    - Prepares espresso.
    - Froths milk for cappuccino and latte.
    - Adds syrups and other ingredients as per customer request.

- The waiter serves the drink on a saucer, adds a spoon, sugar, cream, and a glass of water upon request.

- The waiter brings the prepared drinks to the customer. P.S. Only the waiter serves dishes; the manager may take the order or accept payment (but is not obliged to).

- The customer enjoys the drink and leaves a service rating.
- After finishing the drink and giving a rating, the customer leaves the coffee shop.

- The waiter monitors the condition of the tables and the dining area during customer service. If the customer leaves, the waiter approaches the table, clears the used dishes and trash.

- The waiter wipes the table and sets it up again for the next customer.

- The manager monitors inventory:
- a. (Optional, can be done in 1 process (Before the coffee shop opens)) Receives deliveries:
    - i. Checks goods against invoices.
    - ii. May help distribute goods to the warehouse or shelves.
- b. (Optional, can be done in 1 process (Before the coffee shop opens)) Checks the availability of products on shelves:
    - i. Inspects stocks for the need for replenishment.
    - ii. Updates the database with current inventory status.
- c. (Optional, can be done in 1 process (Before the coffee shop opens)): If a shortage of products is found, sends a notification to the supplier to order.

3 As the coffee shop closes

Barista

- Closes the coffee machine and turns off other equipment.
- Checks for remaining ingredients and stores them in the warehouse or storage.
- Cleans the workstation.

Waiter

- Clears tables of dishes and trash.
- Transfers dirty dishes to the washing area.
- Washes dishes.
- Mops the floors.
- Cleans toilets and replenishes paper.
- Baristas and waiters prepare to leave and ensure all equipment is turned off.

Manager

- Checks cash reports for the day.
- Calculates revenue and profit for the day.
- Ensures all data is saved, and the coffee shop is ready to open the next day.
- Closes the coffee shop and locks the doors.

Table

Process Role	**Activities**
Manager	- Checks equipment condition and calls for repairs if needed. - Prepares the cash register. - Places orders for additional products. - Signs documents for received orders. Greets customers and offers tables. - Takes customer orders (optional). Processes payments and issues receipts. Monitors inventory and orders products if necessary. - Checks cash reports, calculates revenue and profit. - Ensures data is saved and the coffee shop is ready for the next day. - Closes the coffee shop and locks the doors.
Barista	- Prepares workstation and cleans coffee equipment. - Checks ingredient inventory and notifies the manager if reordering is needed. - Prepares coffee drinks (grinding beans, brewing, frothing milk, adding ingredients). Brews tea and prepares hot chocolate. Closes the coffee machine and turns off equipment. - Checks and stores remaining ingredients. Cleans the workstation. - Prepares to leave and ensures equipment is turned off.
Waiter	- Checks the dining area and clears tables. - Checks for clean dishes and utensils.

	- Helps unload delivery trucks (optional). - Greets customers and escorts them to tables. - Takes customer orders. - Serves drinks and food. - Monitors table conditions and clears used dishes. - Wipes tables and sets them up for new customers. - Clears tables of dishes and trash. - Transfers dirty dishes to the washing area. - Washes dishes. - Mops floors. - Cleans toilets and replenishes paper. - Prepares to leave and ensures equipment is turned off.
Customer	- Enters the coffee shop and approaches the cashier. - Decides whether to sit at a table or order directly at the counter. - Places an order and may ask for recommendations or modifications. - Pays for the order (either upfront or after finishing). - Enjoys their drink and food. - Provides feedback or leaves a service rating (optional). - Leaves the coffee shop.

Table 1 - Describes the roles and their assigned activities

Overview process model

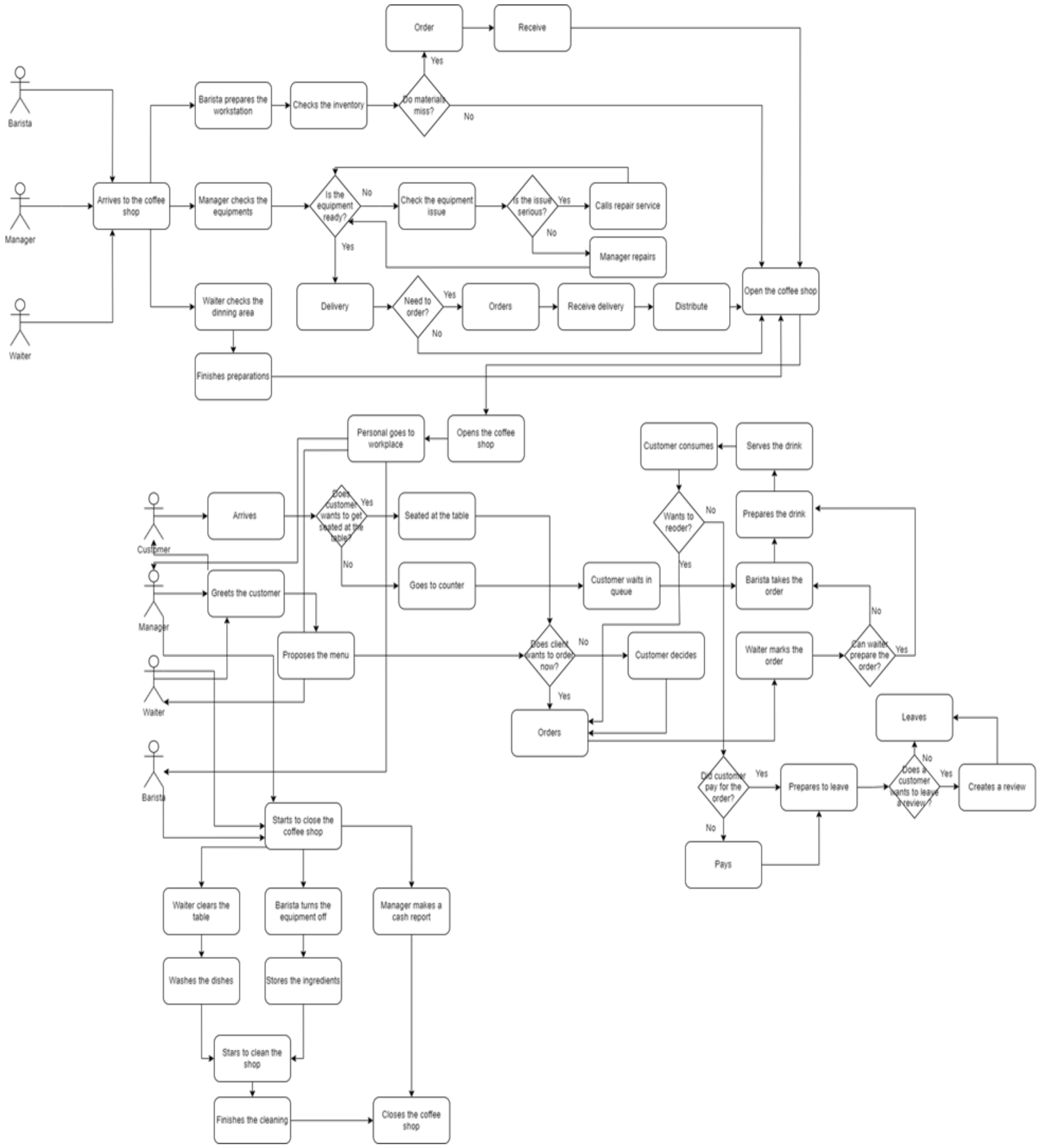

Figure 1 - Overall process diagram

Process model as a Petri net

Reachability graphs

It is possible to create reachability graphs for two of the 3 processes, but it is impossible to create one for the second process because the Petri net is unconstrained due to the ability to add visitors as many times as desired.

Figure 2 - Closing caffe Process

Figure 2 visualizes the closing process of a coffee shop, with three key roles involved: Barista, Waiter, and Manager. Each role has specific tasks to complete, contributing to the overall goal of closing the shop. The Barista handles actions like shutting down equipment, storing ingredients, and washing dishes. The Waiter focuses on clearing tables, while the Manager oversees the entire process, checks reports, and ensures everything is completed correctly. The graph demonstrates the parallel execution of tasks. Multiple actions can happen simultaneously, like storing ingredients, clearing tables, and shutting down equipment. The graph captures the dependencies and the order in which tasks must be completed, reflecting the real-world dynamics of closing a shop. The graph meticulously represents all possible paths and states, illustrating the comprehensive closing process. It showcases the interplay between the roles, highlighting how their individual tasks contribute to the ultimate goal of shutting down the coffee shop.

I could also attach a picture of the reachability graph of the opening process, but it's very big, and on pdf/doc wouldn't be visible or readable at all. All 2 pictures can be found in attachments with this documentation.

Petri net

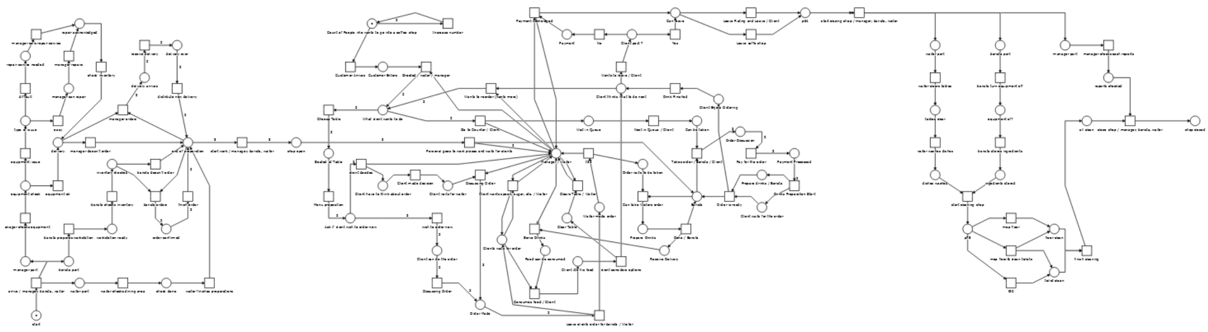

Figure 3 - Whole Petri Net

Figure 3 shows the whole Petri Net of our caffe. For convenience, I will divide them into 3 pictures (of the process) to make it clearer what I am describing

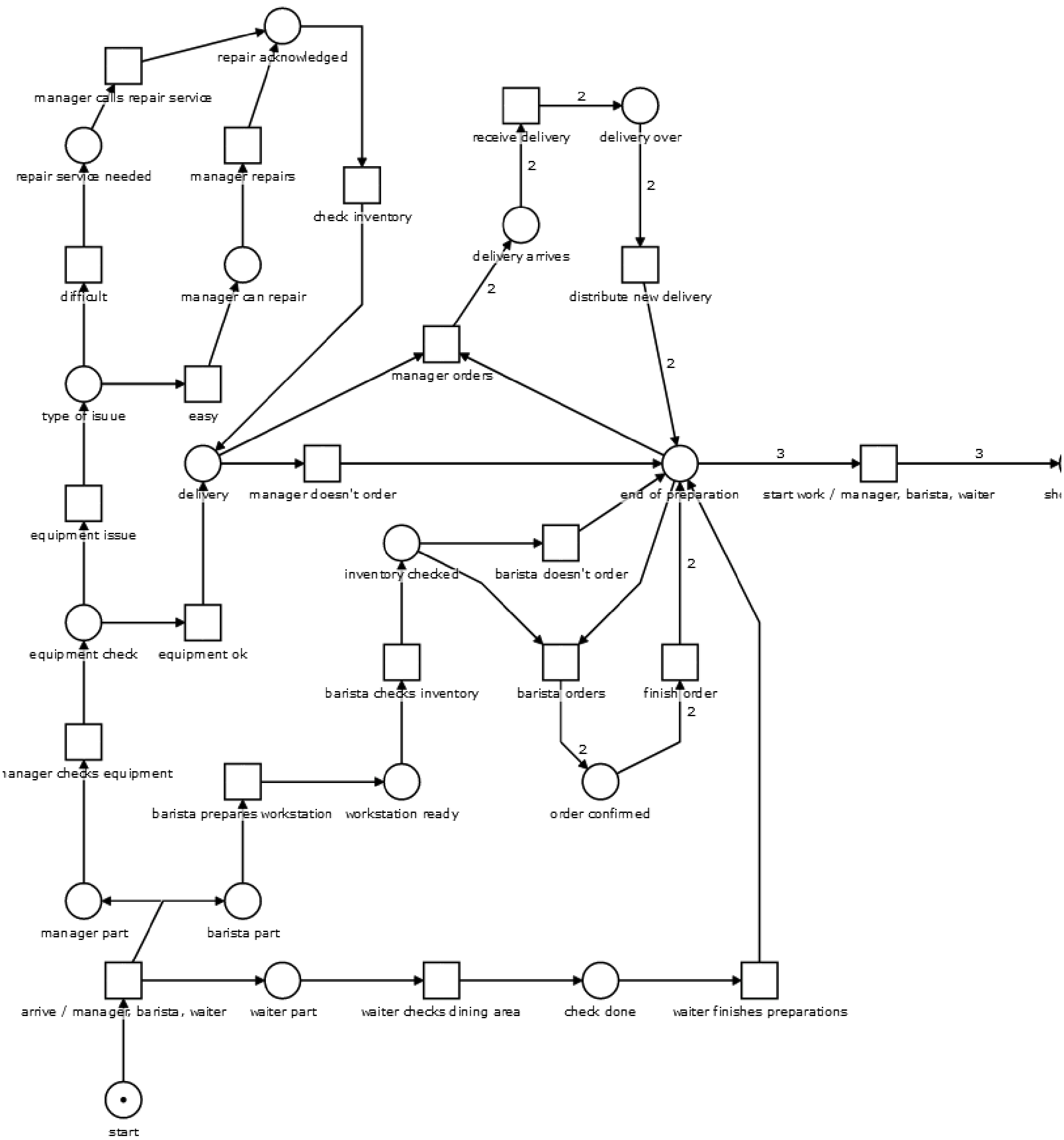

Figure 4 - Caffe Opening

This Petri Net depicts the preparation phase of a coffee shop, focusing on essential tasks like equipment checks, inventory verification, and readying the dining area. It illustrates the interplay between staff roles, showcasing how their individual tasks contribute to the overall readiness of the shop for opening. Also key things in that PN is ordering process, when barista says manager that they have to order, in this state barista waits for manager, and 2 moment is getting ordered staff, it possible can that the order made earlier, for example 2 days ago is here, in that process manager and one more worker, have to participate. And after all preparations are done, the coffee shop can be opened.

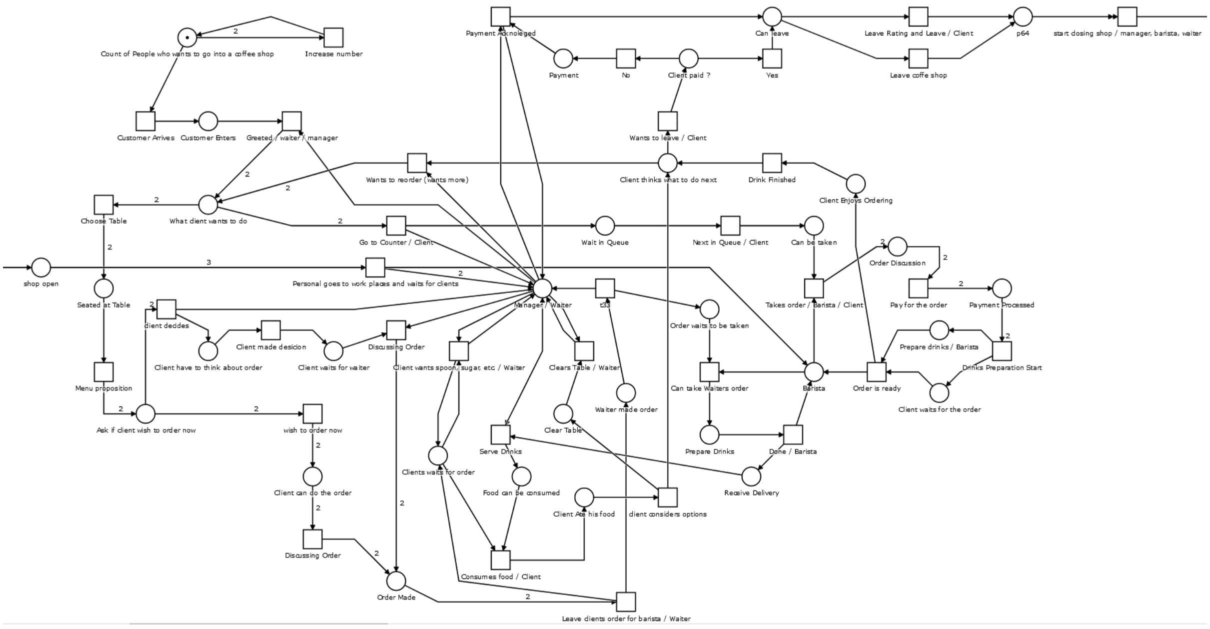

Figure 5 - Full Time Working day

This model simulates the bustling activity of a typical day in a coffee shop, capturing the arrival and ordering processes of multiple customers, along with the dynamic interactions between customers and staff (barista, waiter, manager). It highlights the concurrency of various activities and the complex interdependencies in serving multiple clients. Strength of this PN - opportunity to create a number of clients, before the whole process, this pn simulates real life workload of staff, in our caffe only 3 workers and if one there are none of them for example to get the order, the client have to wait in the queue, etc. Same for client, we left an option for the customer to order something after they had already eaten.

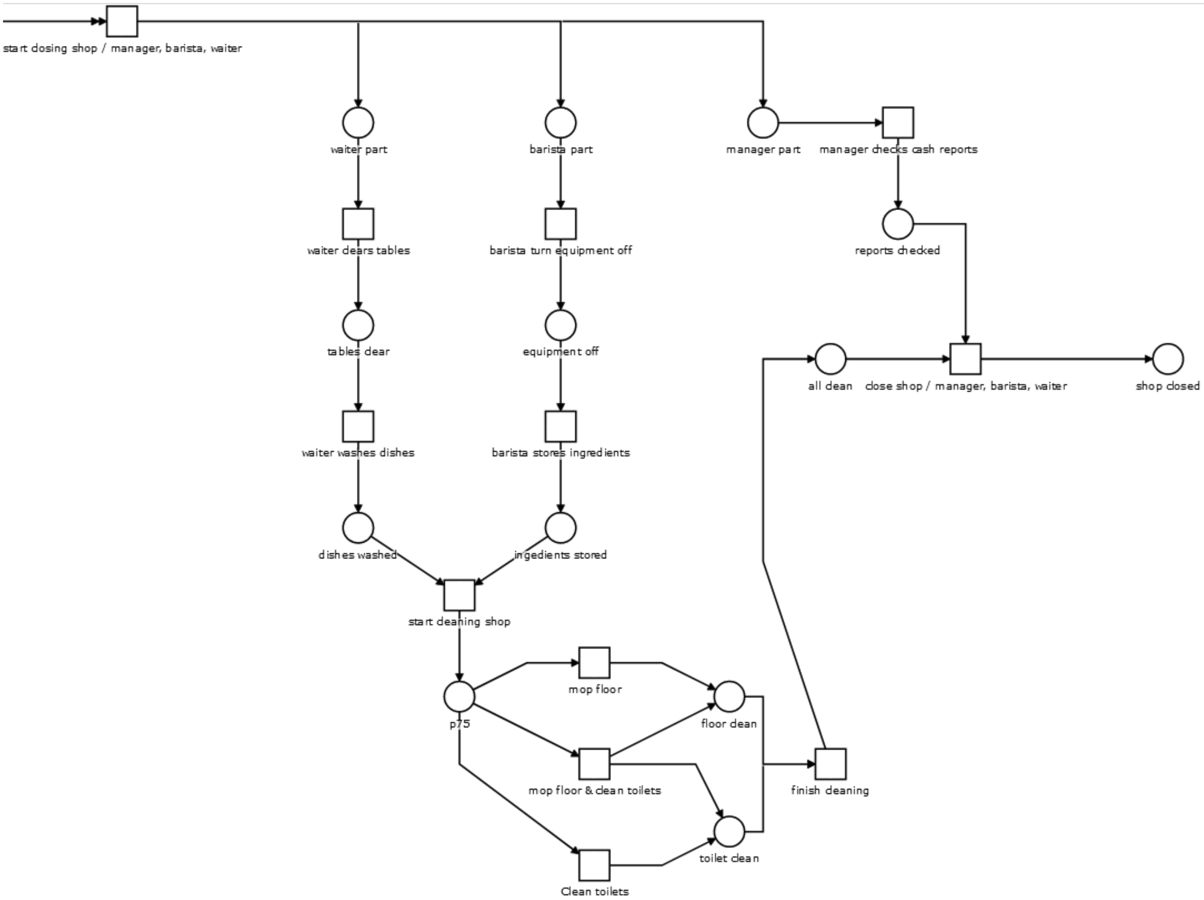

Figure 6 - Caffe Closing

This Petri Net models a streamlined closing process where the barista, waiter, and manager have a clear division of tasks, leading to a systematic shutdown of the coffee shop. It emphasizes the sequential completion of tasks, ensuring a structured and organized closing procedure.

Implementation of a process-driven application

The first step is to deploy the process to the process server.

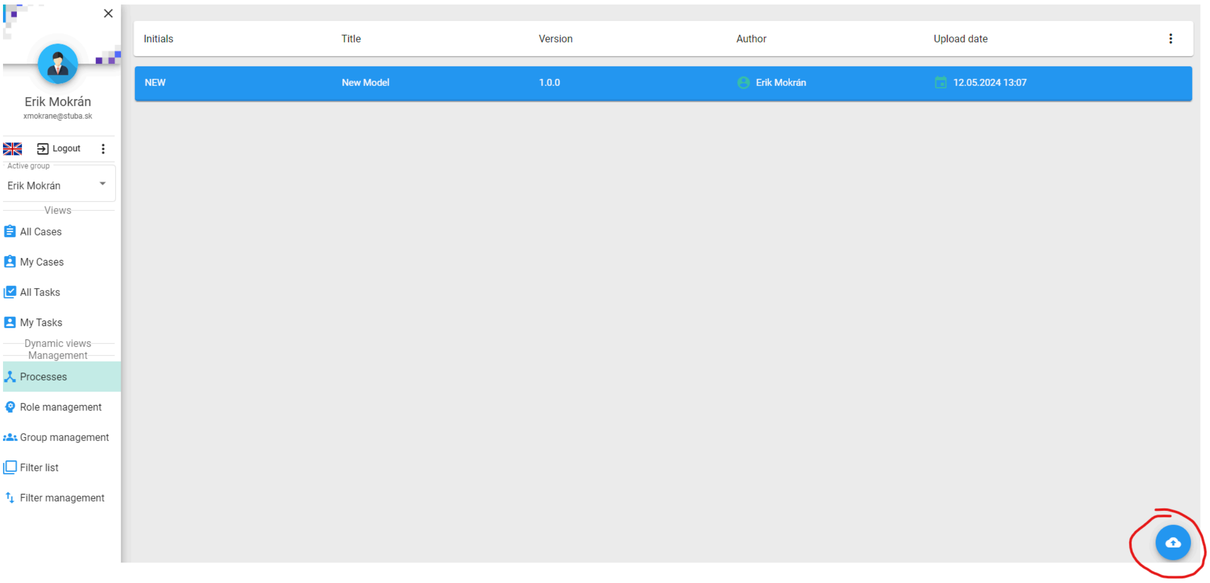

Figure 7 - Deployment of the process

We then assign the roles as we have defined them. Where we assign roles to a given account. For the sake of simplicity, we will assign all roles to one account.

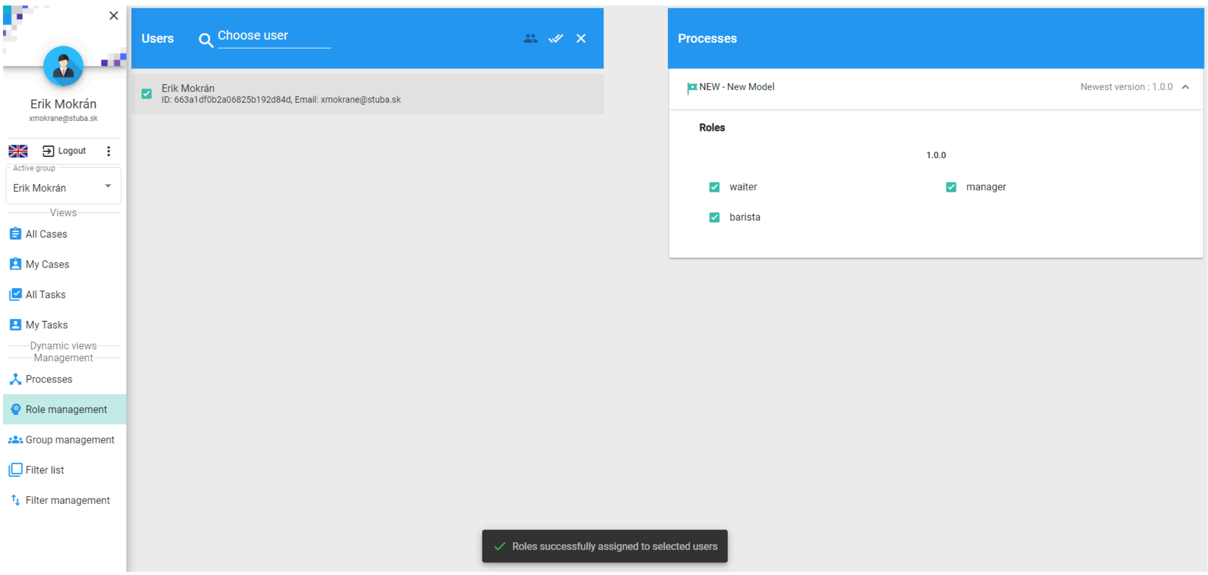

Figure 8 - Creating an instance of a model.

The next step is to instantiate the model. In which we will perform the tasks defined by the roles.Three screens were created for the processes - for opening the coffee shop, for orders and for closing the café.

If the manager discovers that something is missing from the warehouse, he has the option of ordering things to the warehouse.

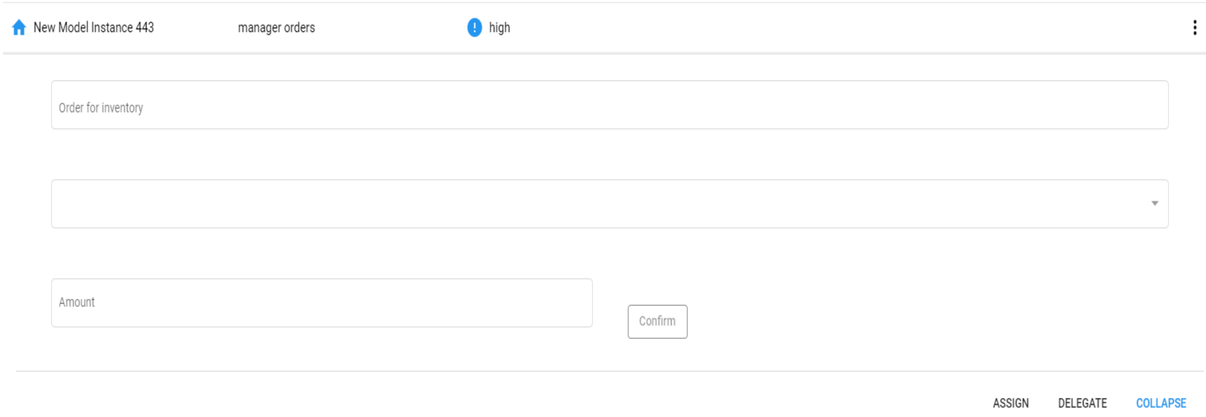

Figure 9 - Order screen for manager to order goods

The screenshot below shows the conditions for opening a coffee shop and whether those conditions are met. We then press the confirmation button, which initializes the opening of the coffee shop.

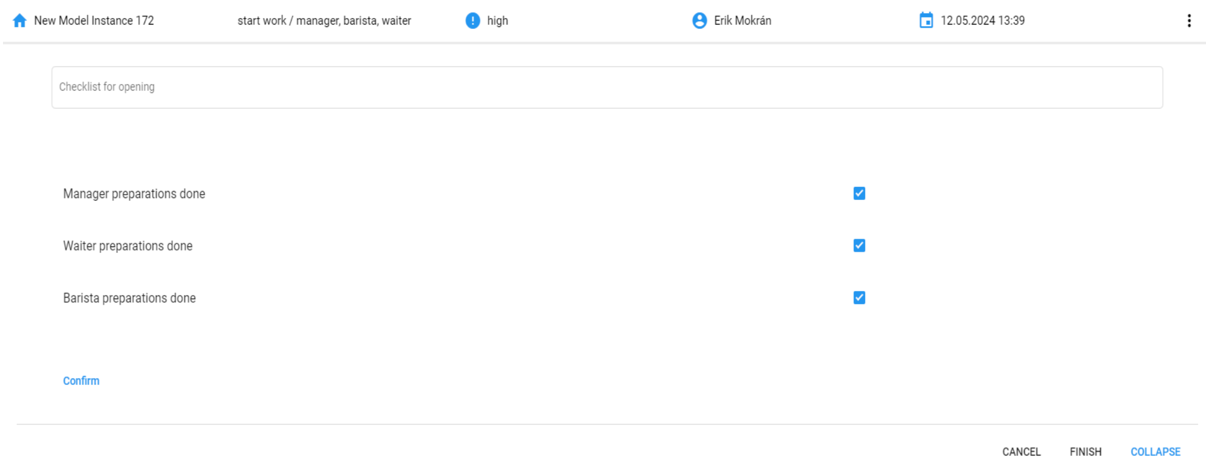

Figure 10 - First screen for opening the shop checklist.

The next screen is the order screen, which is used for the waiter to write down what the customer is ordering. The waiter fills the screen and the order is sent to the system.

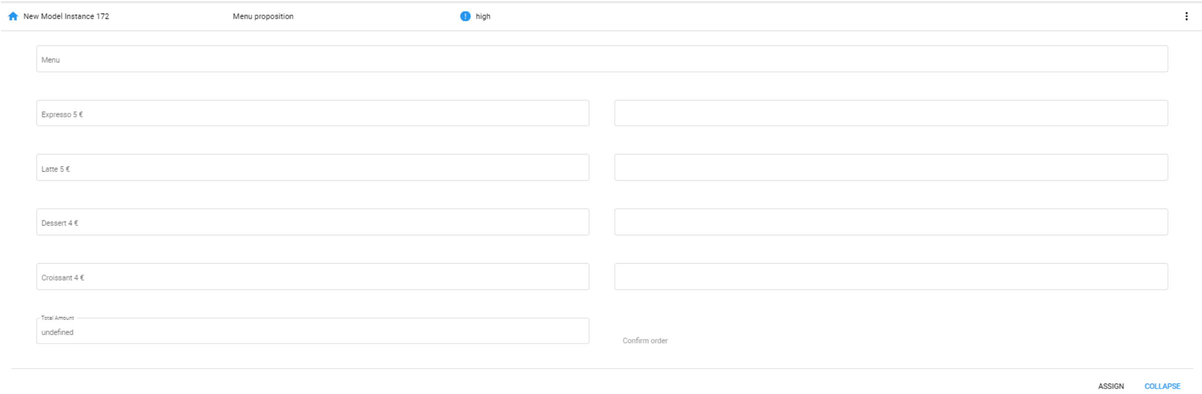

Figure 11 - Screen for waiter to create an order.

The last screen is used to check the conditions used to close the coffee shop.

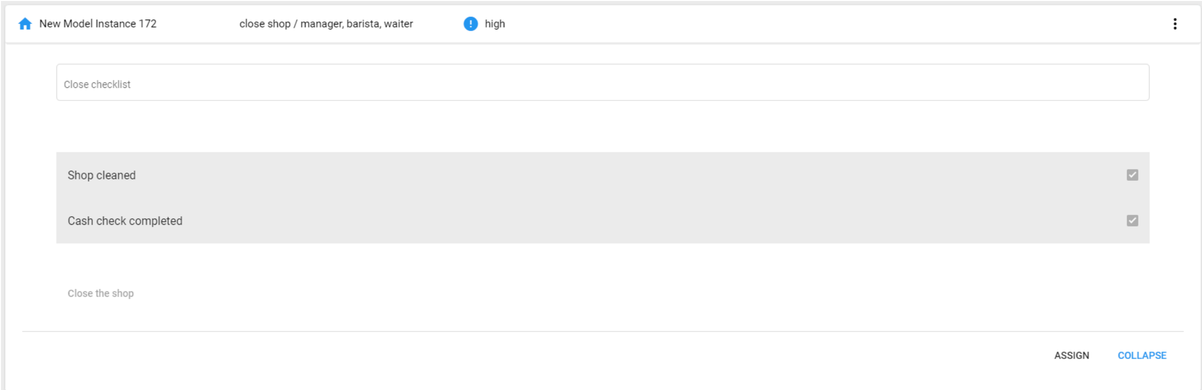

Figure 12 - Screen for closing the coffee shop.

Process model in BPMN

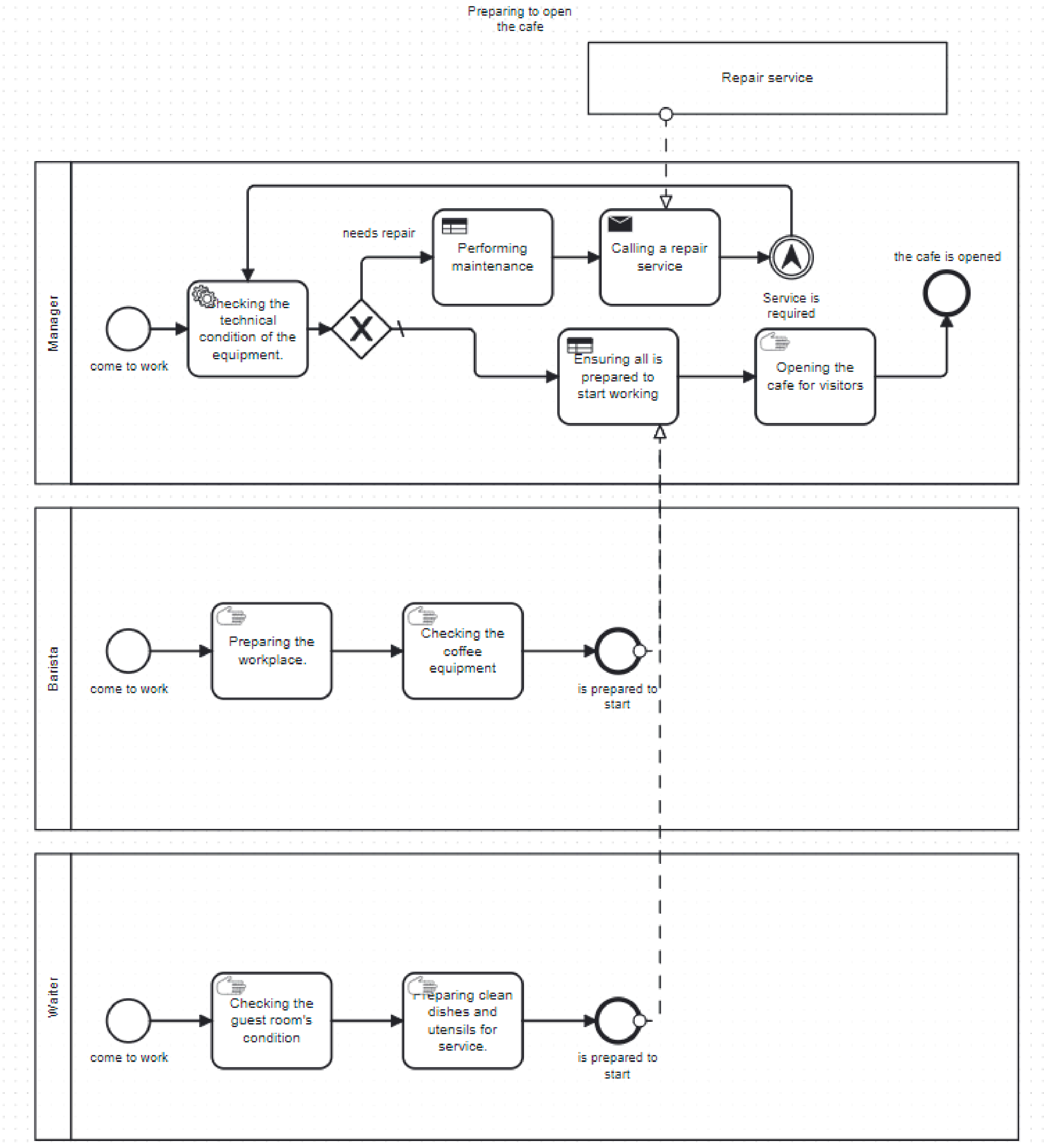

Figure 13 - Model for preparing the coffee shop before opening

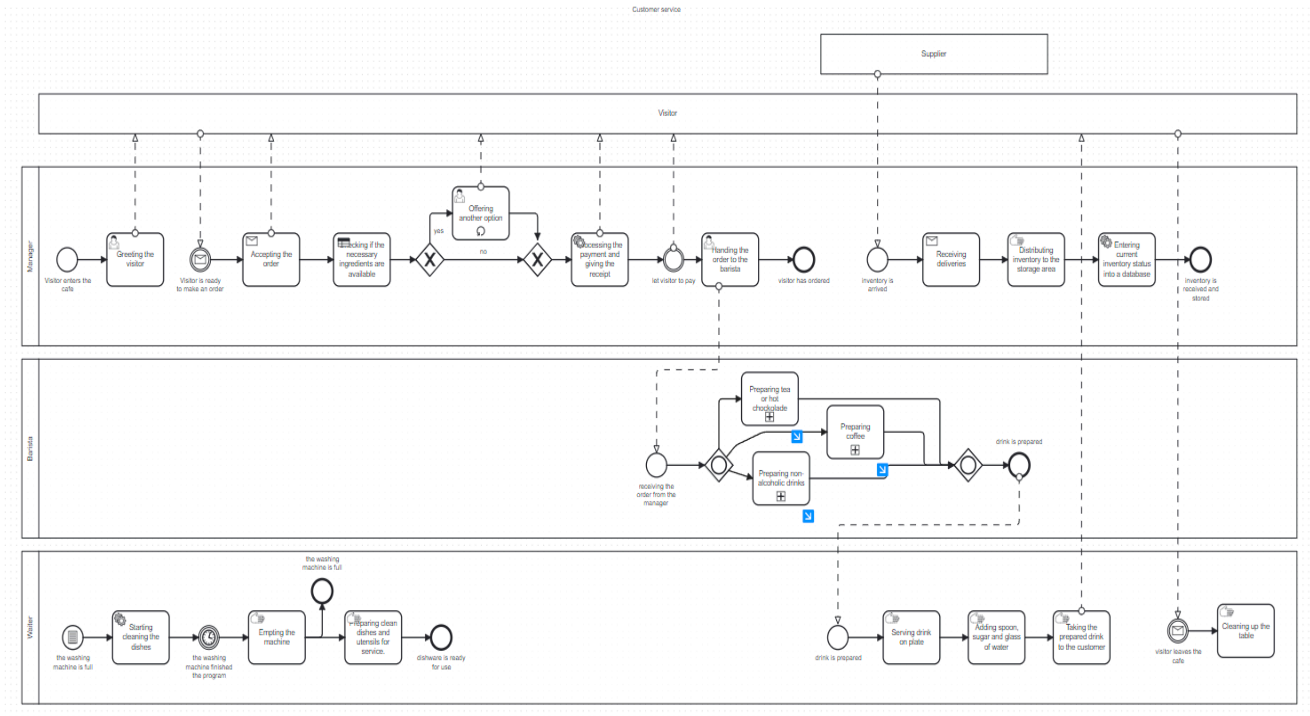

Figure 14 - Describes the whole process of a customer service during the day

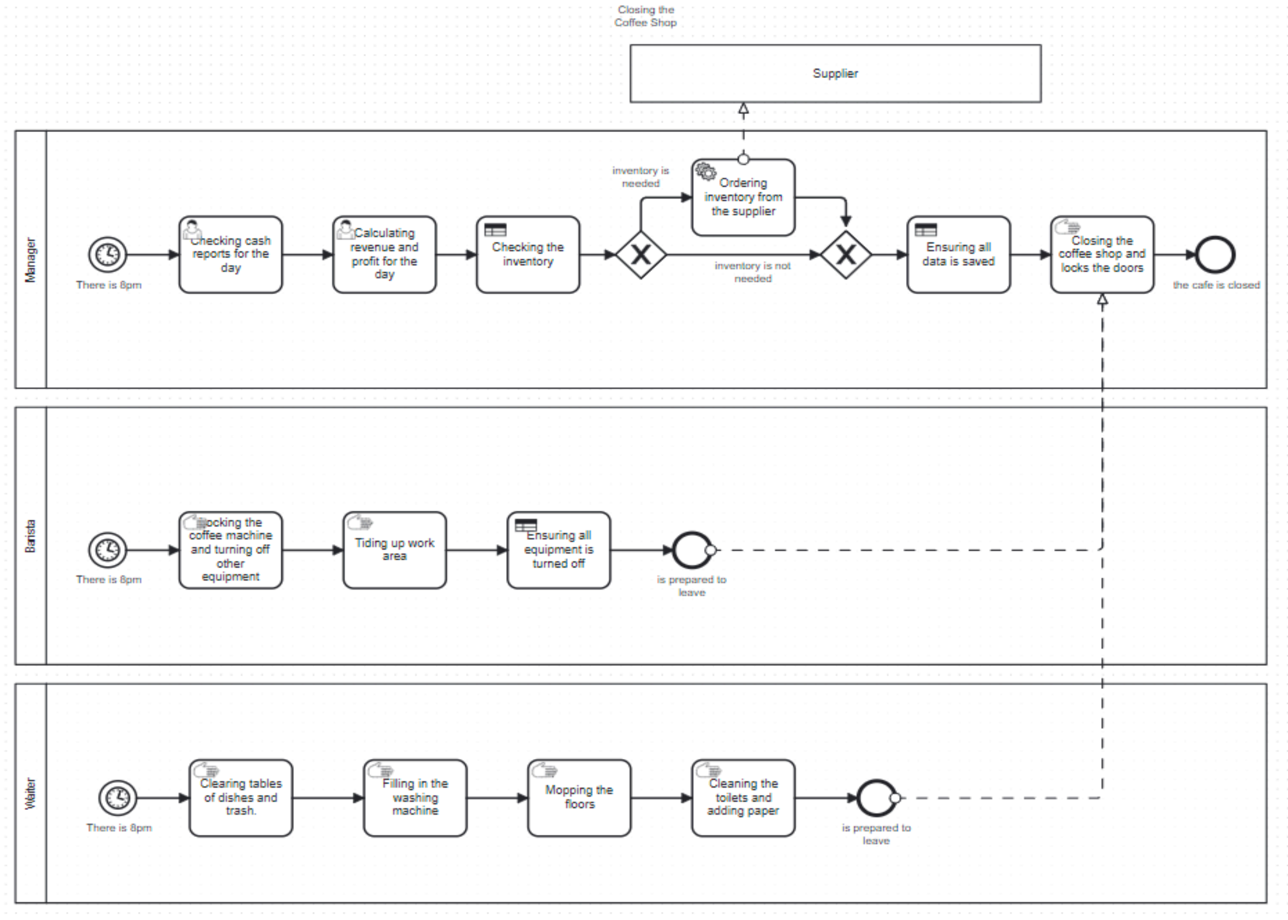

Figure 15 - Describes the closing process